Non-invasive prediction of massive transfusion during surgery using intraoperative hemodynamic monitoring data

Doyun Kwon, Young Mi Jung, Taekyoung Kim, Kwangsoo Kim, Garam Lee,
Dokyoon Kim, Hyung-Chul Lee, Seung-Bo Lee, Seung Mi Lee

Abstract

Introduction: Failure to receive a prompt blood transfusion leads to serious complications after surgery, if massive bleeding occurs during surgery. For the timely preparation of blood products, the prediction of possibility of massive transfusion (MT) is essential to decrease morbidity and mortality. Recently we have reported that intraoperative vital parameters from invasive monitoring can be used for real-time prediction of MT (intra-arterial blood pressure monitoring). However, most surgeries are undergone without intra-arterial blood pressure monitoring and massive bleeding can occur even in these surgeries without invasive monitoring. Therefore, the purpose of this study was to develop a model for predicting MT 10 minutes in advance using non-invasive bio-signal waveforms that change in real-time.

Methods: In this retrospective study, we developed a deep learning-based algorithm (DLA) to predict intraoperative MT before 10 minutes. MT was defined as the transfusion of 3 or more units of red blood cells within an hour. The datasets consisted of 13,835 patients who underwent surgery at Seoul National University Hospital (SNUH) for model development and internal validation, and 528 patients who underwent surgery at Boramae Medical Center (BMC) for external validation, respectively. We used features extracted from plethysmography (collected in 500Hz), non-invasive blood pressure (NIBP), and hematocrit measured during surgery for the construction of DLA.

Results: Among 13,835 patients in SNUH and 528 patients in BMC, 232 patients (1.7%) and 7 patients (1.3%) received MT during surgery, respectively. The areas under the receiver operating characteristic curve (AUROC) of DLA predicting intraoperative MT before 10 minutes was 0.94 (95% CI, 0.910-0.970) in internal validation and 0.93 (95% CI, 0.883-0.972) in external validation, respectively.

Conclusions: The DLA can successfully predict intraoperative MT using non-invasive bio-signal waveforms.

Figure 1. Study design

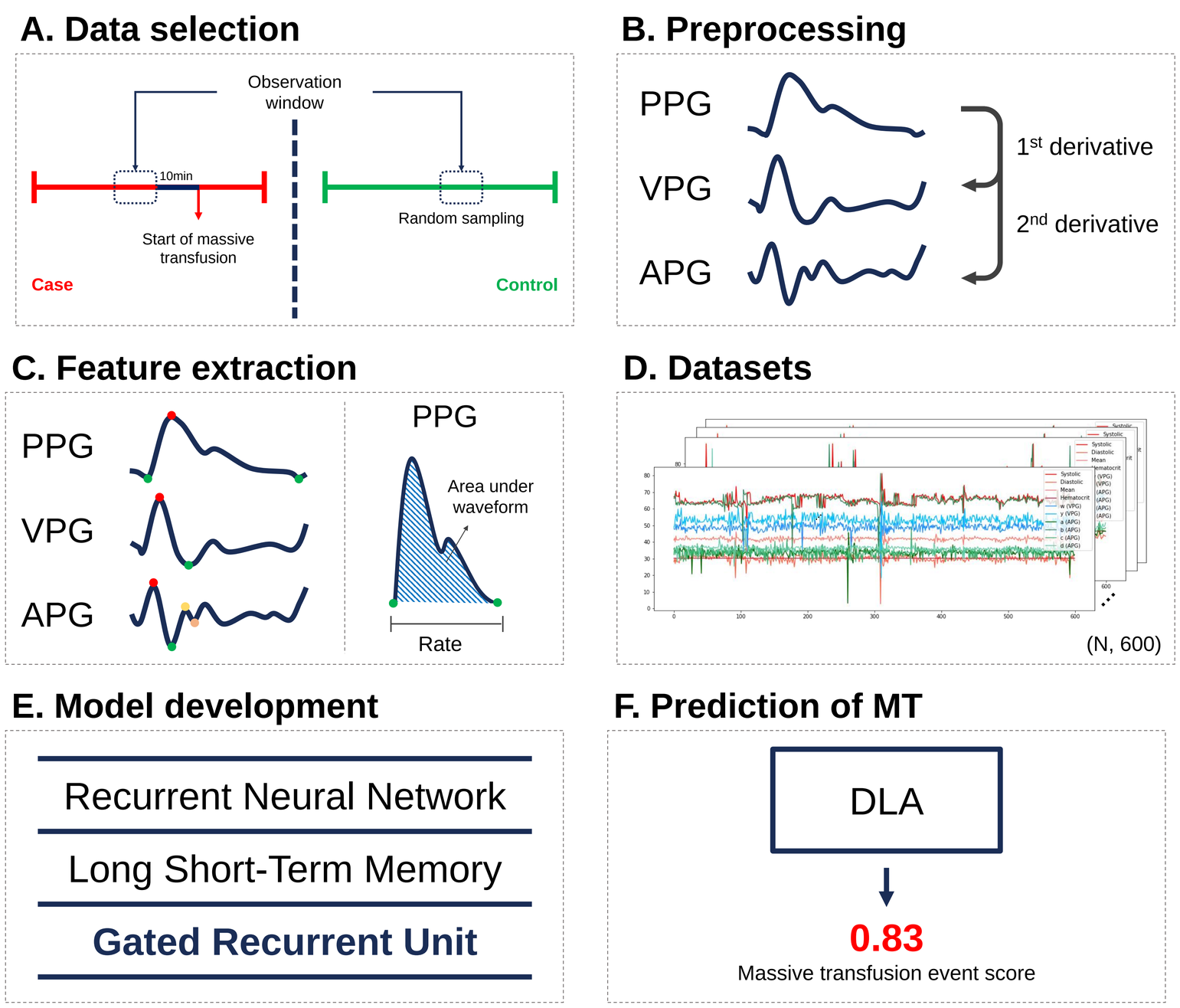
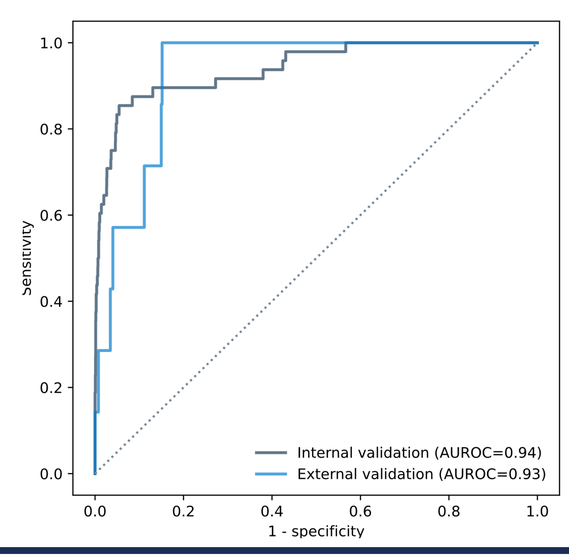


Figure 2. ROC curve



Model : Gated Recurrent Unit
Feature set : Systole, Cycle duration, au_W, hct

Internal validation : **AUROC=0.94**
External validation : **AUROC=0.93**

Table 1. Model performance

Feature set	Internal validation (SNUH)	External validation (BMC)
	AUROC	AUROC
Systole, hct	0.93 (0.909-0.956)	0.89 (0.716-0.992)
Diastole, hct	0.87 (0.826-0.913)	0.89 (0.741-0.981)
Mean, hct	0.92 (0.874-0.960)	0.85 (0.645-0.975)
Cycle duration, hct	0.93 (0.903-0.958)	0.88 (0.716-0.978)
au_W, hct	0.93 (0.901-0.950)	0.87 (0.731-0.990)
Best combination	0.94 (0.910-0.970)	0.93 (0.883-0.972)

Best combination : Systole, Cycle duration, au_W, hct